

Florentin Acker

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EDUCATION

École Polytechnique — Applied Mathematics (M.Sc. level) | 2023 – 2026

France's leading engineering school. Specialising in stochastic analysis, machine learning and scientific computing.

Relevant courses: Statistical Learning, Stochastic Calculus, Evolution Equations, Statistical Modelling (Rosenbaum), Algorithmic Geometry, SciML, Deep Learning.

Lycée Pierre Corneille, Rouen — Classe préparatoire (MPSI/MP*) | 2021 – 2023

Intensive two-year preparation for the competitive entrance exams to top French engineering schools.

RESEARCH & PROFESSIONAL EXPERIENCE

Research Intern — MaLGA, University of Genoa | Apr. 2026 – Aug. 2026

Supervised by Prof. Lorenzo Rosasco.

- Operator learning for stochastic partial differential equations (SPDEs).
- PyTorch implementation with theoretical convergence guarantees.

Algorithm Engineer — Sysnav | Jun. 2025 – Sep. 2025

- Assessed performance of observation information in a Kalman filter.
- Data analysis and mathematical modelling of sensor fusion algorithms.

RESEARCH PROJECTS

Bayesian Deep Learning for Predictive Uncertainty | Mar. 2026

Course project (DL course, École Polytechnique) — co-authored technical report.

- Extended a Bayesian deep learning benchmark (Deep Ensembles, MC-Dropout, HMC) to MNIST classification.
- Implemented a targeted adversarial attack study; analysed aleatoric/epistemic uncertainty decomposition under distribution shift.
- Explored a Jacobian-based epistemic uncertainty proxy (Laplace linearisation), achieving Spearman $\rho = 0.84$ correlation with ensemble scores.

Hawkes Process Modelling of Seismic Activity | Mar. 2026

Course project (Statistical Modelling, École Polytechnique).

- Fitted marked and unmarked Hawkes processes (exponential and power-law kernels) on a 2,961-event USGS seismic catalogue via MLE.
- Model selection via Ogata's time-rescaling theorem; marked exponential Hawkes achieved KS statistic 0.017 ($p \approx 0.39$).

SKILLS

Programming: Python (PyTorch, NumPy, SciPy, scikit-learn), Java, OCaml, LaTeX, web programming.

Mathematics: Stochastic calculus, functional analysis, evolution equations, statistical learning theory, operator theory, complex analysis.

Languages: French (native), English (upper-intermediate), German (intermediate).